

Department of Computer Science and Engineering, SVNIT, Surat
B.Tech. - II (CSE), Semester III, End Semester Examination – November 2025
Course: Computer Organization (CS201)

Date: 28 November 2025

Time: 09:15 AM to 12:15 PM

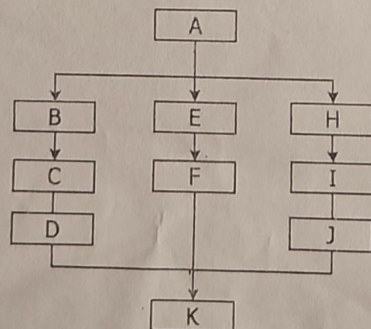
Max. Marks: 100

Instructions:

- Make and state assumptions, as and when required.
- Attempt all questions in the order they are mentioned in the paper.
- Write your details clearly in the answer sheet and on the question paper.

- 1 a. Criticize the following statements: "Using a faster processor chip results in a corresponding increase in performance of a computer even if the main memory speed remains the same". 03
b. In many computers the cache block size is in the range of 32 to 128 bytes. What would be the main advantages and disadvantages of making the size of cache block larger or smaller? 04
c. In general, what are the strategies for exploiting spatial locality and temporal locality? 04
d. Explain with example Structural Dependency, Control Dependency and Data Dependency in a pipelined processor. 06
- 2 a. What is virtual memory? Why it is so called? What are the advantages of having virtual memory system? Explain the mechanism with an example how logical address is converted into physical address? 04
b. Compare and contrast between RISC and CISC. 04
c. Consider the following analogy for the concept of caching. A serviceman comes to a house to repair the heating system. He carries a toolbox that contains a number of tools that he has used recently in similar jobs. He uses these tools repeatedly until he reaches appoint where other tools are needed. It is likely that he has the required tools in his truck outside the house. But, if the needed tools are not in the truck, he must go to his shop to get them.
Suppose we argue that the toolbox, the truck, and the shop correspond to the L1 cache, L2 cache, and the main memory of a computer. How good is this analogy? Discuss its correct and incorrect features. 04
d. Discuss with example basic compiler techniques for exposing Instruction Level Parallelism (ILP). 04

3
6



Above diagram shows a microinstruction sequence that needs to be followed to execute a user program. Two bits from instruction register are used to represent the appropriate branch taken to execute the user program. E.g., when $B_5B_6 = 00$, the branch starting with B will be used. Similarly, when these bits will be 10 (11), the branch associated with E (H) will be executed.